

# Simulating and Testing the Biomarker-by-Drug Interaction in the Hybrid Design: Where We Are

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**Author.** pmsimstats team. **Scope.** The recommended data-generating process (DGP), the recommended analysis, and the evidence for each. Details: docs/36 to docs/38, docs/44 to docs/49, and paper 01.

## 1. The data-generating process

**Starting point.** Hendrickson et al. (2020; “RH”) simulate the score as baseline minus three Gompertz responses: drug (BR), expectancy (PB) and natural course (TV). The biomarker is bound to the drug response as a *correlation*  $c_{bm}$  in a joint multivariate normal, a parameterization new to the N-of-1 literature to our knowledge. Two defects follow from the published implementation.

| Defect in RH’s code                                                                 | Consequence                                              | Evidence                                                                             | Fix                                                                                                                   |
|-------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Biomarker coupled at full strength wherever any drug effect remains (step rule)     | interaction vanishes at any positive half-life           | Hybrid, $N = 70$ : power 0.66 to 0.08 at $t_{1/2} = 0.1$ (RH’s code, 250 replicates) | <b>graded coupling:</b> $c_{bm}$ on drug, $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug, 0 before exposure; power 0.62 to 0.68 |
| Compound symmetry allows $c_{bm}$ only below 0.256; larger values silently repaired | RH’s $c_{bm} = 0.6$ results describe a different process | 20 of 54 matrices repaired                                                           | <b>separable AR(1)</b> ( $\rho = 0.7$ ): ceiling 0.48; <b>stop, do not repair</b>                                     |
| Carryover recursion compounds the decay                                             | off-drug means too small                                 | code inspection                                                                      | <b>anchored carryover</b> from the last on-drug value                                                                 |

**How we got there.** First, closed-form power for a paired-difference statistic showed that power depends on carryover only through the *coupling gap* (on minus off coupling). The step rule sets that gap to 0; graded coupling lets it fall smoothly, and provably monotonically (docs/37). Next, the eight-line fix restored power in RH’s own code. Then the separable form was shown to be valid for every visit schedule, to give an exact ceiling as a sum of per-switch costs, and to match the random-intercept-plus-AR(1) analysis model (docs/35).

**Key insight.** Conditional on the biomarker, covariance binding is mean moderation with a smaller residual variance. How much carryover costs therefore depends on whether the interaction fades with the drug effect, not on where it is bound. Mean binding with the same fading profile loses power almost identically: 0.454 against 0.473 at one week.

### Recommended DGP.

- **Core:** graded coupling, separable AR(1) with  $\rho = 0.7$  and  $c_1 = 0.2$ , anchored carryover, a stop on infeasible matrices.
- **Re-exposure:** the drug-response curve restarts at zero, consistent with re-titration of prazosin.
- **Carryover range:** pharmacological half-life 0.1 to 0.2 week.
- **Options:** prognostic biomarker couplings (natural course, placebo, baseline severity), phase-specific measurement error, Weibull decay, dropout.

Mean binding, or the drug-sensitivity model of docs/38 (not yet implemented), covers effects above the ceiling.

## 2. The analysis

**How we got there.** Three rounds of simulation, each on identical trials, with size judged before power:

1. **Covariance study** (docs/45, 378,000 fits). Kenward-Roger corrects an unstructured covariance; CR2 corrects a parsimonious one. Neither substitutes for the other.

2. **Pre-registered stress test** (docs/46, 39 cells, 26,500 trials, 11 analyses). It varied correlation structure,  $N$ , carryover, decay form, dropout and measurement error, and added a *prognostic* biomarker.
3. **Post hoc arms** on the same trials: phase-specific variances with the expectancy terms (complete), and AIC selection with the expectancy terms (running).

The decisive finding concerns analyses that keep baseline as an outcome row. They share one biomarker slope between baseline and the off-drug visits. If the biomarker also predicts the natural course or the placebo response, part of that association is reported as a drug interaction (docs/48, docs/49).

| Analysis (all random intercept + CAR(1) residuals unless stated) | Size, standard cells       | Size, prognostic cells | Power, AR(1)<br>$c_{bm} = 0.3$ | Power, CS<br>$c_{bm} = 0.25$ |
|------------------------------------------------------------------|----------------------------|------------------------|--------------------------------|------------------------------|
| Published (random intercept, model-based)                        | 0.069 (0.081 at $N = 35$ ) | 0.053 to 0.159         | 0.398                          | 0.596                        |
| Unstructured, Kenward-Roger                                      | 0.057 (0.087 at $N = 35$ ) | 0.195 to 0.416         | 0.534                          | 0.618                        |
| S: fixed half-life 0.5, CR2                                      | 0.048                      | 0.049 to 0.100         | 0.514                          | 0.554                        |
| DS: S with AIC-selected half-life                                | 0.053                      | 0.058 to 0.115         | 0.486                          | 0.480                        |
| C1: S with phase-specific variances                              | 0.053                      | 0.173 to 0.382         | 0.578                          | 0.660                        |
| <b>F1: S plus expectancy terms</b>                               | <b>0.051</b>               | <b>0.040 to 0.047</b>  | <b>0.364</b>                   | <b>0.522</b>                 |
| <b>De + bmc:De</b>                                               |                            |                        |                                |                              |
| G1: F1 with phase-specific variances (post hoc)                  | 0.049                      | 0.042 to 0.055         | 0.354                          | 0.546                        |

*Size: null rejection, nominal 0.05 (standard cells pooled over 11,000 trials; band 0.046 to 0.054). Power is size-adjusted, 500 trials per cell. "Prognostic": the biomarker correlated 0.3 with the natural course (constant or growing) or with the placebo response. A baseline-severity scenario is running.*

#### Reading the table.

- **Phase-specific variances (C1)** are the most powerful, but only because they weight the shared baseline slope heavily. Once the baseline is freed (G1), the power advantage disappears and the robustness is kept. G1 is a tie with F1, slightly better under compound symmetry and under prognostic alternatives.
- **AIC selection (DS)** holds its size and estimates the carryover half-life sensibly: it picks 0 82% of the time when there is none, and 1 or 2 weeks 86% of the time when it is 1. It costs about 0.03 power against a pre-specified 0.5 week. It does not protect against a prognostic biomarker by itself.
- **The expectancy terms are what make an analysis robust.** Their cost is about 30% of power under strong serial correlation, and small under compound symmetry.

#### Recommended analysis.

- **Primary:** F1. Random intercept with corCAR1 residuals, CR2 standard errors, linear time, baseline as an outcome row, the exposure-decayed drug indicator, and De + bmc:De.
- **Carryover half-life:** pre-specified at 0.5 week. If face validity is preferred, AIC selection over {0, 0.25, 0.5, 1, 2} (ML for selection, REML refit) at about 0.03 power, pending the F1+AIC run.
- **Sensitivity analyses:** S (biomarker assumed purely predictive), the freed-baseline version, and the binary drug indicator. Disagreement between F1 and S is evidence of a prognostic biomarker.
- **Sample size:** plan it for F1, stating the serial-correlation assumption.

### 3. Open items

- **Pending runs:** F1 with AIC selection, and the baseline-severity cells.
- **Package changes:** move the refinements into the package, and remove the silent repair in its sampler.
- **Calibration:** calibrate effect sizes to prazosin data. Blood pressure's placebo-arm association was null in 35 participants (Raskind et al. 2016), too few to rule out a prognostic effect.
- **Confirmation:** a confirmatory run for any post hoc choice (G1, AIC).